MILITARY SPECIFICATION

RESISTOR, FIXED, FILM, HIGH RELIABILITY
GENERAL SPECIFICATION FOR

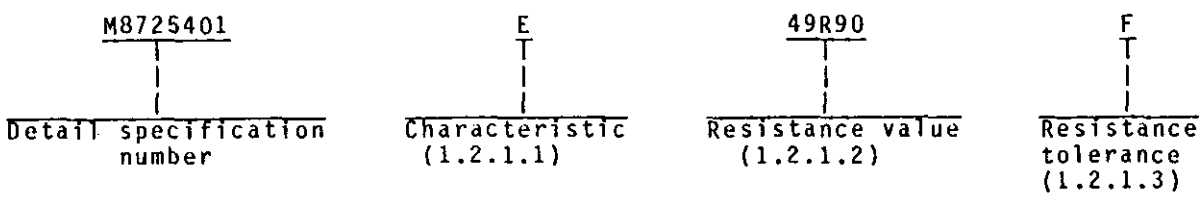
This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for hermetically sealed (see 3.12) and nonhermetically sealed (see 3.12.1) high reliability, fixed, film resistors for space, missile and other high reliability applications. The prerequisite to qualifying to this specification shall be based on current qualification to "S" failure rate level for detail specifications one through four and "R" failure rate level for detail specification five on a similar MIL-R-55182 resistor style. The resistance temperature characteristic will be limited to ±5 ppm/°C for the RSC90 style and ±25 ppm°C for the remaining styles. Resistance tolerances will be limited to ±0.1 percent and ±1.0 percent. Part or Identifying Number (PIN) is a new term encompassing previous terms used in specifications such as part number, type designator, and identification number.

1.2 Classification.

1.2.1 Part or Identifying Number (PIN). Resistors specified herein shall be identified by the PIN which shall consist of the basic number, associated detail specification, and a coded number. Each associated detail specification covers a different resistor style. The number shall be coded to provide information concerning resistor characteristic, resistance value, and tolerance. The PIN shall be in the following form with a coded number derived as indicated:



Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: SD/ALM, P.O. Box 92960, Worldway Postal Center, Los Angeles, CA 90009-2960, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1.1 Characteristic. The characteristic is identified by a single letter that signifies the level of stability through various electrical and environmental tests and operating characteristics. Characteristic C, E, H, J, or Y is available with the stability and tolerance as noted (see table I).

1.2.1.2 Resistance. The nominal resistance expressed in ohms is identified by five characters, consisting of four digits and a letter. This is used simultaneously as a decimal point and a multiplier. For resistance values:

a. Greater than or equal to 10 ohms but equal to or less than 988 ohms, the letter "R" is used to represent a decimal point.

b. Greater than or equal to 1,000 ohms but equal to or less than 988 kilohms, the letter "K" is used to represent a decimal point.

c. Greater than or equal to one megohm, the letter "M" is used to represent a decimal point.

All digits preceding and following the letter (R, K, or M) of the group represent significant figures. Minimum and maximum resistance values shall be as specified herein. The standard values for every decade shall follow the sequence specified in table II, herein for resistance tolerances B and F. The resistance values for tolerance B may be any value within the limits specified in 3.1, but it is preferred that values be chosen from the "B" column of table II, herein. The resistance value designations are shown in table III.

1.2.1.3 Resistance tolerance. The resistance tolerance, when applicable, shall be identified by a single letter in accordance with table I.

TABLE I. Characteristic.

Hermetically sealed	Nonhermetically sealed	Resistance temperature characteristics (referred to 25°C)	Tolerance	
			Symbol	±%
		(Parts/million/°C)		
	Y	±5	B	0.1
E	J	±25	B	0.1
C	H	±50	F	1.0

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TABLE II. Standard resistance values for the 10 to 100 decades.

Resistance tolerance 1/											
B .1	F 1.0	B .1	F 1.0	B .1	F 1.0	B .1	F 1.0	B .1	F 1.0	B .1	F 1.0
10.0	10.0	14.7	14.7	21.5	21.5	31.6	31.6	46.4	46.4	68.1	68.1
10.1	---	14.9	---	21.8	---	32.0	---	47.0	---	69.0	---
10.2	10.2	15.0	15.0	22.1	22.1	32.4	32.4	47.5	47.5	69.8	69.8
10.4	---	15.2	---	22.3	---	32.8	---	48.1	---	70.6	---
10.5	10.5	15.4	15.4	22.6	22.6	33.2	33.2	48.7	48.7	71.5	71.5
10.6	---	15.6	---	22.9	---	33.6	---	49.3	---	72.3	---
10.7	10.7	15.8	15.8	23.2	23.2	34.0	34.0	49.9	49.9	73.2	73.2
10.9	---	16.0	---	23.4	---	34.4	---	50.5	---	74.1	---
11.0	11.0	16.2	16.2	23.7	23.7	34.8	34.8	51.1	51.1	75.0	75.0
11.1	---	16.4	---	24.0	---	35.2	---	51.7	---	75.9	---
11.3	11.3	16.5	16.5	24.3	24.3	35.7	35.7	52.3	52.3	76.8	76.8
11.4	---	16.7	---	24.6	---	36.1	---	53.0	---	77.7	---
11.5	11.5	16.9	16.9	24.9	24.9	36.5	36.5	53.6	53.6	78.7	78.7
11.7	---	17.2	---	25.2	---	37.0	---	54.2	---	79.6	---
11.8	11.8	17.4	17.4	25.5	25.5	37.4	37.4	54.9	54.9	80.6	80.6
12.0	---	17.6	---	25.8	---	37.9	---	55.6	---	81.6	---
12.1	12.1	17.8	17.8	26.1	26.1	38.3	38.3	56.2	56.2	82.5	82.5
12.3	---	18.0	---	26.4	---	36.8	---	56.9	---	83.5	---
12.4	12.4	18.2	18.2	26.7	26.7	39.2	39.2	57.6	57.6	84.5	84.5
12.6	---	18.4	---	27.1	---	39.7	---	58.3	---	85.6	---
12.7	12.7	18.7	18.7	27.4	27.4	40.2	40.2	59.0	59.0	86.6	86.6
12.9	---	18.9	---	27.7	---	40.7	---	59.7	---	87.6	---
13.0	13.0	19.1	19.1	28.0	28.0	41.2	41.2	60.4	60.4	88.7	88.7
13.2	---	19.3	---	28.4	---	41.7	---	61.2	---	89.8	---
13.3	13.3	19.6	19.6	28.7	28.7	42.2	42.2	61.9	61.9	90.9	90.9
13.5	---	19.8	---	29.1	---	42.7	---	62.6	---	92.0	---
13.7	13.7	20.0	20.0	29.4	29.4	43.2	43.2	63.4	63.4	93.1	93.1
13.8	---	20.3	---	29.8	---	43.7	---	64.2	---	94.2	---
14.0	14.0	20.5	20.5	30.1	30.1	44.2	44.2	64.9	64.9	95.3	95.3
14.2	---	20.8	---	30.5	---	44.8	---	65.7	---	96.5	---
14.3	14.3	21.0	21.0	30.9	30.9	45.3	45.3	66.5	66.5	97.6	97.6
14.5	---	21.3	---	31.2	---	45.9	---	67.3	---	98.8	---

1/ The resistance values for tolerance B may be of any value, but it is preferred that the values be chosen from the tolerance B values in this table.

TABLE III. Designation of resistance values.

Designation	Resistance
10R00 to 98R80	10 to 98.8 ohms included
100R0 to 988R0	100 to 988 ohms included
1K000 to 9K880	1 to 9.88 kilohms included
10K00 to 98K80	10 to 98.8 kilohms included
100K0 to 988K0	100 to 988 kilohms included
1M000 to 9M880	1 to 9.88 megohms included

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-14256	-	Flux, Soldering, Liquid (Rosin Base).
MIL-R-39032	-	Resistor, Packaging of.
MIL-R-55182	-	Resistor, Fixed, Film, Established Reliability.
MIL-R-87254/1	-	Resistor, Fixed, Film, High Reliability, Style RSC50.
MIL-R-87254/2	-	Resistor, Fixed, Film, High Reliability, Style RSC55.
MIL-R-87254/3	-	Resistor, Fixed, Film, High Reliability, Style RSC60.
MIL-R-87254/4	-	Resistor, Fixed, Film, High Reliability, Style RSC65.
MIL-R-87254/5	-	Resistor, Fixed, Film, High Reliability, Style RSC90.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-790	-	Reliability Assurance Program for Electronic Parts Specifications.
MIL-STD-883	-	Test Methods and Procedures for Microelectronics.
MIL-STD-1276	-	Leads for Electronic Component Parts.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenye, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Associated detail specifications. The individual item requirements shall be as specified herein and in accordance with the applicable detail specification. In the event of any conflict between the requirements of this specification and the detail specifications, the latter shall govern (see 6.1).

3.2 Qualification. Resistors furnished under this specification shall be products which are authorized by the qualifying activity for listing on both qualified products lists at the time of award of contract (see 4.4 and 6.2) for MIL-R-55182 (with minimum failure rate 'R' for RNC90's and 'S' for the rest) and herein. The manufacturer shall obtain certification from the qualifying activity that the reliability assurance requirements of 4.1.1 have been met and are being maintained. Unless acquired from a manufacturer and approved for listing, on the qualified products list, parts furnished under this specification shall not be considered as having met the requirements of this specification. Distributors should not be used for distribution of parts. In addition, as a prerequisite for qualification, the manufacturer shall demonstrate to the qualifying activity that the requirements for outgassing have been met (see 3.22).

3.2.1 Reliability and baseline documentation. Resistors supplied to this specification shall conform to the requirements of MIL-STD-790 and the requirements specified herein. Prior to granting of qualification approval, a baseline document shall be prepared by the manufacturer and submitted to the qualifying activity for approval.

3.3 Material. The material shall be as specified herein and in accordance with the baseline document (see 3.2.1). In addition, materials used in the manufacturing of the resistor shall meet the outgassing requirements (see 3.22).

3.4 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified (see 3.1) and in accordance with the baseline document (see 3.2.1). Resistors furnished under characteristic E shall consist of a film-type resistance element, protected from exposure to humidity by a hermetically sealed enclosure (3.12). Characteristics J and Y resistors shall consist of a film-type resistance element, but need not be hermetically sealed (see 3.12.1).

3.4.1 Metallizing of shell ends. Metallizing, when used, shall be uniform around the periphery of the ends of the shell, shall be well bonded to the base material, and shall provide a good seal with the mating part.

3.4.2 Films. Films shall be uniformly deposited. The film shall be free of blisters, thin spots, areas inadequately bonded to the core, discolored spots, or other blemishes likely to cause flaking or a nonuniform ribbon when spiraled (helixed). Where used, spiraling shall occupy no less than 70 percent of resistor-element actual length. The resistor-element actual length shall be defined as the nominal distance between terminal bands less 3/64-inch.

3.4.3 Terminal leads. Terminal leads shall be made of a solid conductor of the length and diameter specified. The lead composition and final finish shall be specified by C31, C32, or C52 (maximum thickness of 200 microinches is not applicable) in accordance with MIL-STD-1276. In addition manufacturers may supply type N22 lead finish for characteristic E. The leads shall be solderable in accordance with MIL-STD-202, method 208, beyond the maximum specified clean lead-to-clean lead dimension.

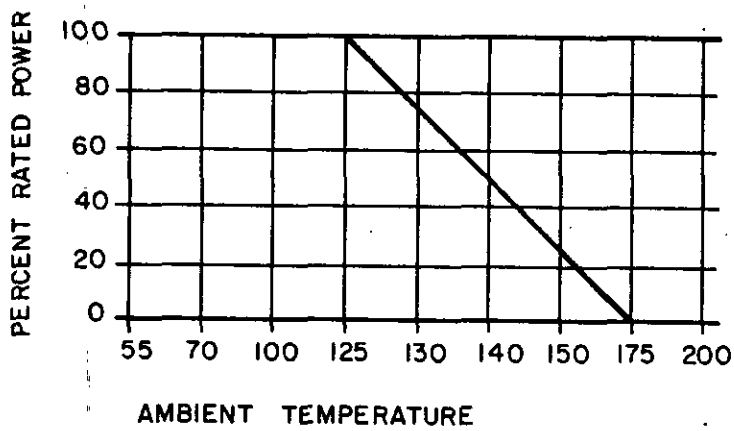
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3.4.4 Flux. The manufacturer shall verify by certification that only noncorrosive solder fluxes shall be used unless it can be shown that corrosive elements have been removed satisfactorily after soldering. Noncorrosive fluxes shall comply with the requirements for resistivity of water extract and effect on vacuum-deposited copper specified in MIL-F-14256. If cored solders are used, the proportion of flux to solder by volume shall be between one percent and three percent.

3.4.5 End caps. When end caps are used in construction of the resistor, the misalignment of the cap with respect to the core shall not exceed 3°C.

3.4.6 Insulation. Insulation material surrounding the resistor body shall extend around the entire shoulder of each end cap toward the axis of the lead attachments. Complete back wall coverage is optional (not applicable to RSC90).

3.5 Power rating. Resistors shall have a power rating based on continuous full-load operation at an ambient temperature of 125°C. This power rating is dependent on the ability of resistors to meet the life requirements specified in 3.22. For temperatures in excess of those specified above, the load shall be derated in accordance with figure 1.



NOTE: This curve indicates the percentage of nominal wattage to be applied at temperatures higher than 125°C. However, at no time shall the applied voltage exceed the maximum for each style.

FIGURE 1. Derating curves for high ambient temperatures.

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3.6 Voltage rating. Resistors shall have a rated direct-current (dc) continuous working voltage or an approximate sine-wave root-mean-square (rms) continuous working voltage at commercial-line frequency and waveform corresponding to the power rating, as determined from the following formula:

$$E = PR$$

Where: E = rated dc or rms continuous working voltage.
P = power rating (see 3.1).
R = nominal resistance.

In no case shall the rated dc or rms continuous working voltage be greater than the maximum value (see 3.1).

3.7 Thermal shock. When resistors are tested as specified in 4.7.2, there shall be no evidence of mechanical damage; the change in resistance shall not exceed (±0.1 percent +0.01 ohm or ±3 standard deviation, whichever is less) for group A inspection, thermal shock, and overload tests combined (see 3.8); and ±0.2 percent +0.01 ohm, for group C inspection.

3.8 Overload. When resistors are tested as specified in 4.7.3, there shall be no evidence of arcing, burning, or charring; the change in resistance shall not exceed (±0.1 percent +0.01 ohm or ±3 standard deviation, whichever is less for thermal shock and overload tests combined (see 3.7).

3.9 Power conditioning. When resistors are tested as specified in 4.7.2.1, there shall be no evidence of mechanical damage and the change in resistance shall not exceed ±0.2 percent +0.01 ohm or ±3 standard deviation whichever is less.

3.10 DC resistance. When resistors are tested as specified in 4.7.4, the dc resistance shall be within the specified tolerance of the nominal resistance (see 1.2.1.3).

3.11 Hermetic seal (applicable to characteristic C and E). For the purposes of this specification, a hermetically sealed resistor is one in which the resistance element is contained within a sealed enclosure of ceramic, glass, or metal, or combinations thereof, where sealing is accomplished by material fusion, welding, brazing, or soldering. The resistors supplied under characteristic E shall be capable of passing the seal test specified in 4.7.5 or of meeting a leak rate requirement of not more than 1×10^{-8} cubic centimeters (cc) per second, when determined by method 112 of MIL-STD-202, procedure IV, and there shall be no evidence of dye emanating from the resistor when tested in accordance with method 1014 of MIL-STD-883, condition D.

3.11.1 Nonhermetic seal (applicable to characteristics H, J, and Y). For the purposes of this specification, a nonhermetically sealed resistor is one not conforming in full to the requirements of 3.11. A qualified resistor which meets all requirements of 3.11 may be furnished to meet these characteristics; however, the resistor must continue to meet all requirements of characteristic E. Characteristic J and Y resistors shall not be furnished against characteristic E requirements.

3.12 Solderability. When resistors are tested as specified in 4.7.6, they shall meet the criteria for wire-lead terminal evaluation in the test method.

3.13 Resistance to solvents. When resistors are tested as specified in 4.7.7, there shall be no evidence of mechanical damage and the markings shall remain legible.

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3.14 Resistance-temperature characteristic. When resistors are tested as specified in 4.7.8, the resistance-temperature characteristic, at each of the temperatures specified in table IX, referred to room temperature, shall not exceed the value specified in table II for the applicable characteristic.

3.15 Terminal strength. When resistors are tested as specified in 4.7.9, there shall be no evidence of breaking or loosening of terminals from the resistor form, or chipping of coating, or other evidence of mechanical damage. The change in resistance shall not exceed ±0.2 percent +0.01 ohm.

3.16 Dielectric withstanding voltage. When resistors are tested as specified in 4.7.10, there shall be no evidence of flashover, mechanical damage, arcing, or insulation breakdown. The change in resistance shall not exceed ±(0.15 percent +0.01 ohm).

3.17 Insulation resistance. When resistors are tested as specified in 4.7.11, the insulation resistance shall be not less than 1×10^{10} ohms.

3.18 Resistance to soldering heat. When resistors are tested as specified in 4.7.12, there shall be no evidence of mechanical damage. The change in resistance shall not exceed ±(0.1 percent +0.01 ohm).

3.19 Moisture resistance. When resistors are tested as specified in 4.7.13, there shall be no evidence of mechanical damage. The change in resistance for characteristics H, J, and Y shall not exceed ±(0.4 percent +0.01 ohm); for characteristic C and E, the change in resistance shall not exceed ±(0.2 percent +0.01 ohm). In addition, the dielectric withstanding voltage shall be as specified in 3.17, and the insulation resistance shall be 1×10^{8} ohms, minimum.

3.20 Resistance noise. When tested in accordance with 4.7.14 the third harmonic index (THI) shall be screened to accept only product within a ±3 standard deviation.

3.21 Destructive physical analysis. When examined as specified in 4.7.15 resistors shall meet the requirements in appendix B of this document.

3.22 Outgassing. When examined as specified in 4.7.16, the samples shall meet the following requirements:

a. Total mass loss (Tml) shall not exceed 1 percent.

b. Volatile condensable material (VCM) shall not exceed 0.1 percent.

3.23 Marking.

3.23.1 Full marking. Resistors shall be marked with the part number designation, "JAN" marking, date code, source code, and manufacturer's production lot code (see 6.1). Date and source code shall be in accordance with MIL-STD-1285. At the option of the manufacturer, the part number designation may appear on two lines. In this event, the part number designation shall be divided between the characteristic letter and the first digit of the resistance value. The following is an example of the complete marking:

12345	- Source code
7242J	- Date code and JAN marking
M87254**	- Military specification and detail specification
******	- Characteristic, resistance value, and tolerance
*****	- Manufacturer's production lot code

The date code shall be the date of the final assembly operation for the production lot as defined in 6.1. The common manufacturing record shall include the same date code as that placed on the parts covered by the record.

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3.23.2 Minimum marking. When the physical size of the resistor style precludes the marking of all the above information, the minimum marking required shall be as specified in the detail specification (see 3.1). Marking shall remain legible at the end of all tests. In those cases where full marking requirements are not on the resistor body, the full marking shall be marked on the unit package.

3.23.3 JAN and J marking. The United States Government has adopted, and is exercising legitimate control over the certification marks "JAN" and "J", respectively, to indicate that items so marked or identified are manufactured to, and meet all the requirements of military specifications. Accordingly, items acquired to, and meeting all of the criteria specified herein and in applicable specifications shall bear the certification mark "JAN" except that items too small to bear the certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed immediately before the part number except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN" or "J" may be located on the first line above or below the part number. Items furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein or in applicable specifications shall not bear "JAN" or "J". In the event an item fails to meet the requirements of this specification and the applicable specification sheets or associated detail specifications, the manufacturer shall remove the "JAN" or the "J" from the sample tested and also from all items represented by the sample. The "JAN" or "J" certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration Number 504,860 for the certification mark "JAN".

3.23.4 Use of conductive inks. Conductive inks shall not be used to coat the body of the resistor as preparation for marking, or used for marking of resistors.

3.24 Workmanship. Resistors shall be processed in such a manner as to be uniform in quality and shall be free from any defects that will affect life, serviceability, or appearance.

3.24.1 Soldering. Electrical connections shall be mechanically secure before soldering and electrically continuous after soldering.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Reliability assurance program. A reliability assurance program shall be established and maintained in accordance with MIL-STD-790. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification.

4.1.2 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).

b. Verification of qualification (see 4.5).

c. Quality conformance inspection (see 4.6).

4.3 Inspection conditions and precautions.

4.3.1 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.3.2 Precautions. Adequate precautions shall be taken during inspection to prevent condensation of moisture on resistors, except during the moisture-resistance temperature-cycling tests. Precautions shall also be taken to prevent damage by heat when soldering resistor leads to terminals.

4.4 Qualification inspection. Initial qualification shall be based upon the manufacturers qualification to a minimum 'R' failure rate for RNC90 only and 'S' failure rate for the remaining styles, of the equivalent MIL-R-55182 resistor being offered for listing on the qualified products list of this specification. Additionally, the manufacturer shall pass the qualifying activity audit (see appendix A).

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4.5 Verification of qualification. Every 6 months the manufacturer shall compile a summary of the results of quality conformance inspections in the form of a verification of qualification report, and forward it to the qualifying activity within 30 days after the end of the reporting period as the basis for continued qualification. If the summary of test results indicates noncompliance with specification requirements, and a corrective action acceptable to the qualifying activity has not been undertaken, the qualifying activity may take action to remove the failing product from the qualified products list. In addition, the manufacturer shall immediately notify within 15 days the qualifying activity whenever the data indicates that the manufacturer has failed to maintain his qualified product. Continuation shall be based on evidence that over the 6-month period the following has been met:

a. The manufacturer has not modified the design of the item.

b. Parts manufactured have met all specification requirements.

c. Lot rejection for group A inspection does not exceed 3 percent, or one lot, whichever is greater.

4.6 Quality conformance inspection. Parts supplied to this specification shall have been manufactured on the same production line as the equivalent qualified MIL-R-55182 resistor.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspection.

4.6.1.1 Inspection lot (group A). An inspection lot shall be of one design as specified in the baseline documentation. As a minimum requirement, the lot shall consist of resistors of the same style, characteristic, resistance value, voltage rating and tolerance, and shall use slugs from a single deposition/metallization. All resistors in the lot shall have been started, processed, assembled and tested as a group on the same manufacturing/test equipment, to the same baseline/revision, and identified with the same lot date code designation. Traceability of all raw materials used in the lot shall be maintained. In addition, the deposition/metallization shall be restricted to the following:

a. The slugs produced shall be those for a single resistor style.

b. The slugs shall be processed in a single continuous metallization run, on the same equipment.

c. The raw materials used in the metallization process shall be from the same raw material lots.

4.6.1.1.1 Rework. Rework is not allowed on any operation except marking and lead rework.

4.6.1.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table IV, and shall be made on the same set of sample units, in the order shown.

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TABLE IV. Group A inspection.

Examination or test	Requirement paragraph	Method paragraph	AQL (percent defective and LQ limiting quality)	
			Major	Minor
Subgroup 1 (100% inspection)				
Thermal shock 1/	3.7	4.7.2		
Overload 1/	3.8	4.7.3		
Power conditioning	3.9	4.7.2.1		
DC resistance 2/	3.10	4.7.4	100%	
Subgroup 2 (100% inspection)				
Resistance noise	3.20	4.7.14		
Subgroup 3				
Visual and mechanical 3/	3.1, 3.3, 3.4, 3.4.1 3.4.2, 3.4.3 3.4.5, 3.4.6 3.24, and 3.24.1	4.7.1	32 samples 0 failures	
Marking and workmanship	3.24 and 3.24.1			
Subgroup 4 (100% inspection)				
Hermetic seal test 4/	3.11	4.7.5		
Subgroup 5				
Destructive physical anaylsis	3.21	4.7.15		
hermetically sealed			5 samples	
nonhermetically sealed			8 samples 0 failures	

1/ At the manufacturer's option, the determination of resistance change may be by any method which is within the accuracy requirements of this specification. The determination of resistance change shall be made upon completion of the overload test; the allowable change in resistance for the combined tests shall be specified (see 3.7).

2/ Resistors shall meet the specified initial resistance tolerance after being subjected to the preceding tests. The resistance measurement made upon completion of the overload test may be used if a measurement has been made which can, without conversion, be directly related to nominal resistance value and tolerance.

3/ At the manufacturer's option, marking inspection may be performed after group B inspection; however, if this is done, no defects shall be permitted. Marking shall remain legible at end of all tests. If the sample fails the 0.65 AQL, the lot shall be 100 percent screened.

4/ Hermetic seal test applicable to characteristic C and E.

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4.6.1.2.1 Sampling plan. Subgroups 1, 2, and 4 tests shall be performed on 100 percent of the product supplied under this specification. Resistors that are not of resistance tolerance or which experience a change in resistance greater than that permitted for the test of this subgroup shall be removed from the lot. Lots having more than 3 percent total rejects, or one resistor, whichever is greater, due to exceeding the specified resistance change limit, shall not be furnished on the contract or purchase order.

4.6.1.2.1.1 Subgroups 1, 2, and 4 acceptance. Only lots having not more than 3 percent rejects, or one resistor, whichever is greater, as a result of subgroups 1 and 4 tests combined shall be furnished on orders. In addition any lot having more than 3 percent rejects as a result of subgroup 2 tests shall not be furnished on orders.

4.6.1.2.2 Manufacturer's production inspection. If the manufacturer performs tests equal to or more stringent than those specified in subgroup 1, table IV, as the final step of this production process, group A, subgroup 1 inspection may be waived and the data resulting from the manufacturer's production tests may be used instead. Authority to waive the subgroup 1 inspection shall be granted by the qualifying activity only. The following criteria must be complied with:

a. Tests conducted by the manufacturer during production shall be clearly identical to or more stringent than those specified for subgroup 1. Test conditions shall be equal to or more stringent than those specified for subgroup 1.

b. Manufacturer subjects 100 percent of the product supplied under this specification to his production tests.

c. The parameters measured and the failure criteria shall be the same or more stringent than those specified herein.

d. The lot rejection criteria is the same or more stringent than that specified herein.

e. The manufacturer shall make available all information concerning the test procedures and instrumentation used in his production tests. This data shall be provided as part of the evaluation required for MIL-STD-790 and shall be documented in the baseline. The manufacturer shall also make available to the Government all records of all detailed test data resulting from production tests.

f. Once approved the manufacturer shall not change the test procedures or criteria without prior notification and concurrence by the qualifying activity and subsequent change in the baseline.

4.6.1.2.3 Rejected lots (subgroup 3). Rejected inspection lots may be resubmitted for government acceptance only if the manufacturer performs 100 percent inspection on resistors of the lot for those characteristics which were defective and resulted in rejection of lot, removes all defective units and resubmits the lot for quality conformance inspection. Resubmitted lots shall be kept separate from new lots, and shall be clearly identified as resubmitted lots. A new sample of parts shall then be randomly selected in accordance with table IV. If one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

4.6.1.3 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table V, in the order shown. They shall be performed on lots which have been subjected to and passed the group A inspection.

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4.6.1.3.1 Sample plan. Group B sampling shall consist of the lot of parts that are of the same style and characteristic and shall represent a period of time not to exceed one week in duration.

4.6.1.3.2 Rejected lots. If an inspection lot is rejected, the manufacturer may rework it in accordance with 4.6.1.1.1 to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

TABLE V. Group B inspection.

Examination or test	Requirement paragraph	Method paragraph	Number of samples	No. of defects allowed
Subgroup 1				
Solderability	3.13	4.7.6	5	0
Dielectric withstanding voltage	3.17	4.7.10		
Insulation resistance	3.18	4.7.11		
Subgroup 1A				
Resistance to solvents	3.14	4.7.7	8	0
Subgroup 2				
Visual and mechanical examination (when applicable) 1/	3.1 & 3.24	4.7.1	5	0
Resistance-temperature characteristic	3.15	4.7.8		
Subgroup 3				
Resistance to soldering heat	3.19	4.7.12	6	0
Moisture resistance	3.20	4.7.13		

1/ Only applicable when marking inspection is not performed in group A inspection. No defects shall be permitted. Marking shall remain legible at end of all tests.

4.6.1.3.3 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on contract or purchase order.

4.6.2 Periodic inspection. Periodic inspection shall consist of group C inspection. Except where the results of this inspection shows noncompliance with the applicable specification requirements, delivery of products that have passed groups A and B inspections shall not be delayed pending the results of the periodic inspection. Group C data shall be submitted to the qualifying activity every six months.

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4.6.2.1 Group C inspection. Group C inspection shall consist of the test specified in table VI, in the order shown, and shall be performed on sample units of each style and characteristic produced during the inspection period covered and shall be selected from lots that have passed groups A and B inspections. The allowable number of defectives shall be as indicated in table VI. The assigned periodic tests shall be initiated within 30 days of the specified periodic inspection.

TABLE VI. Group C inspection.

Examination or test	Requirement paragraph	Method paragraph	Number of samples	No. of defects allowed
Monthly				
Subgroup 1				
Thermal shock (100 cycles)	3.7	4.7.2	50	0
Subgroup 2				
Terminal strength 1/	3.16	4.7.9	6	0

1/ The terminal strength test will be performed on both leads of all samples drawn and shall be exposed to two test conditions.

NOTE: Marking shall remain legible at end of all tests.

4.6.2.1.1 Sampling plan.

4.6.2.1.1.1 Monthly (subgroup I). Each month the tests specified in subgroup I shall be performed on 50 sample units of each style and characteristic on resistance values in the highest decade produced. The samples selected from the highest decade shall be representative of the decade, and the month's production samples selected shall not be from low value end of decade of values chosen.

4.6.2.1.1.2 Monthly (subgroup II). Each month one half of the sample units of any resistance value below critical and one half of the sample units of any resistance value above the critical value of each style and characteristic shall be subjected to the tests of subgroup I, in the order shown.

4.6.2.1.2 Disposition of samples. Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order.

4.6.2.1.3 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall immediately notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the qualifying activity). Groups A and B inspection may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

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4.6.3 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-R-39032.

4.6.4 Retention of data and test samples. The manufacturer shall retain the following data and test samples for twelve years.

a. Group B and C samples

b. DPA samples

c. Lot travelers

d. Lot test information

e. Results of special screens

4.7 Methods of examination and tests.

4.7.1 Visual and mechanical examination. Resistors shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, 3.4.1, 3.4.2, 3.4.3, 3.4.5, 3.4.6, 3.24, and 3.24.1 inclusive).

4.7.2 Thermal shock (see 3.7). Resistors shall be tested in accordance with method 107 of MIL-STD-202. The following details and exceptions shall apply:

a. Special mounting:

(1) Quality conformance inspection-resistors shall be mounted by means other than soldering or may be placed in trays. When trays are used, they must be designed to present a minimum of obstruction to the airstream. In no case shall the fixture prevent the specified ambient temperature from being achieved within 4 minutes after resistors are placed in chamber. One chamber may be used for this test.

b. Measurement before cycling-dc resistance shall be measured as specified in 4.7.4.

c. Test condition:

(1) Quality conformance inspection-test condition F(5 cycles) for group A and F-3(100 cycles) for group C except that the extreme high temperature shall be 150°C +10, -0°C and the extreme low temperature shall be -65°C +0, -10°C. These extreme temperatures shall be achieved within 4 minutes.

d. Measurements after cycling-dc resistance shall be measured as specified in 4.7.4 following test of 4.7.3.

e. Examination after test-resistors shall be examined for evidence of mechanical damage.

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4.7.2.1 Power conditioning (see 3.9). Resistors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply: Resistors shall be mounted as in 4.7.3 for quality conformance inspection. The load applied shall be full rated power for a duration of 168 ±4 hours, and at a test temperaure of 125 ±3°C. The maximum voltage shall be as specified (see 3.1). The voltage applied may be dc or ac. DC resistance as specified in 4.7.4 shall be measured before the test and after the test following a two hour stabilization period of 25°C ±2°C. Resistors shall be visually examined for any evidence of mechanical damage.

4.7.3 Overload (see 3.8).

4.7.3.1 Mounting. Resistors may be mounted in any position and allotted any size space as deemed necessary by the manufacturer. Forced air cooling may be used to maintain a test ambient temperature range of 20°C to 45°C. The average velocity of the forced air, if employed, shall not exceed 500 feet per minute.

4.7.3.2 Procedure. Table VII lists the load which shall be applied, its duration, and the maximum voltage, alternating current (ac) or direct current (dc), which may be applied to the applicable resistor style.

TABLE VII. Overload procedure.

Resistor style 1/	Overload and duration	Maximum voltage (ac or dc)
		Volts
RSC 50, 55, 60	5 x rated power for 1 hour	500
RSC65	4 x rated power for 1 hour	600
RSC90	6.25 x rated power for 5 seconds	450

Resistance as specified in 4.7.4 shall be measured following the tests, and compared with reading taken prior to test of 4.7.2. Resistors shall be examined for evidence of arcing, burning, or charring.

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4.7.4 DC resistance (see 3.10). Resistors shall be tested in accordance with method 303 of MIL-STD-202. The following details and exceptions shall apply:

a. Measuring apparatus: The same measuring instrument shall be used for any one test, but not necessarily for all tests.

b. Limit of error of measuring apparatus: ±(0.1% +0.01 ohm), but not exceeding one-fourth of the resistor tolerance or the resistance-change limit for which the measurement is being made. Manufacturers, at their option, may use apparatus of less accuracy, provided that limits are reduced to fully compensate for accuracy deviation.

c. Test voltage: Measurements of resistance shall be made by using the test voltages specified in table VIII. The test voltage chosen, whether it be the maximum or a lower voltage which would still provide the sensitivity required, shall be applied across the terminals of the resistor. This same voltage shall be used whenever a subsequent resistance measurement is made.

TABLE VIII. DC resistance test voltages.

Resistance, nominal	Maximum test voltage	
	1/2 watt	1/20, 1/10, 1/8, and 1/4 watt
Ohms	Volts	Volts
10 to 98.8 incl	1	1
100 to 988 incl	3	3
1000 to 9800 incl	10	3
10000 to 98800 incl	30	10
100000 or higher	100	30

d. Temperature: The dc resistance test specified in group I of table V shall be performed at 25° ±2°C. For all other tests, unless otherwise specified herein, the temperature at which subsequent and final resistance measurements are made in each test shall be within ±2°C of the temperature at which the initial resistance measurement was made.

4.7.5 Hermetic seal (applicable to characteristic C and E)(see 3.11). Resistors shall be tested in accordance with method 112 of MIL-STD-202, and method 1014 of MIL-STD-883. The following details shall apply:

a. Group C inspection: Test condition C, procedure III or IV of method 112, MIL-STD-202 shall be used (procedure IV will be used as a referee in the event of disagreement), except the gross leak test shall be in accordance with 4.7.5b. The coating (organic or nonorganic) may be removed prior to testing.

b. Group A inspection: Method 1014, condition D of MIL-STD-883.

(1) For transparent envelopes, the chamber shall be evacuated to 28-30 inches of mercury gauge pressure for a minimum of 2 minutes prior to pressurization. The mixing of dye solution with hydraulic fluid is permissible when such fluids are used to achieve chamber pressure above 1,000 psi.

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(2) For nontransparent envelopes, the chamber shall be evacuated as above. After pressurization, resistors shall be thoroughly cleansed of external dye and then rotated about their longitudinal axis for a minimum of 2 minutes at minimum ambient temperature of 80°C. Following rotation, resistors shall be examined for evidence of dye leakage.

4.7.6 Solderability (see 3.12). Resistors shall be tested in accordance with method 208 of MIL-STD-202. The following detail shall apply: Both leads shall be tested.

4.7.7 Resistance to solvents (see 3.13). Resistors shall be tested in accordance with method 215 of MIL-STD-202. The following shall apply:

a. The marked portion of the resistor body shall be brushed.

b. The number of sample units shall be as specified in table V, as applicable.

c. Resistors shall be examined for mechanical damage and legibility of markings.

4.7.8 Resistance-temperature characteristics (see 3.14). Resistors shall be tested in accordance with method 304 of MIL-STD-202. The following detail and exceptions shall apply:

a. Reference temperature: Room ambient temperature.

b. Test temperature: In accordance with table IX.

TABLE IX. Resistance-temperature characteristic.

Sequence	Group B inspection °C
1	Room temperature 1/
2	-55±3
3	Room temperature 1/
4	-----
5	175±3
6	---
7	---

1/ This temperature shall be considered the reference temperature for each of the succeeding temperatures.

c. Accuracy of temperatures measurement: Resistors shall be maintained for 30 to 45 minutes within 1°C at each of the test temperatures listed in table IX. This tolerance shall be maintained at the established test temperatures.

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4.7.9 Terminal strength (see 3.15). Resistors shall be tested in accordance with method 211 of MIL-STD-202 with the following exceptions:

a. Test conditions: A and D (pull test and twist test, respectively).

b. Measurement before test (A): DC resistance as specified in 4.7.4.

c. Method of holding (A): Resistors shall be clamped by one terminal lead.

d. Applied force (A): Two pounds for styles RSC50, RSC55, RSC60, and RSC65.

e. Measurement after test (D): DC resistance as specified in 4.7.4, and examined for evidence of breaking and loosening of terminals and chipping of coating.

4.7.10 Dielectric withstanding voltage (see 3.16).

4.7.10.1 Atmospheric pressure. Resistors shall be tested in accordance with method 301 of MIL-STD-202. The following details and exceptions shall apply:

a. Special preparations: Resistors shall be clamped in the trough of 90° metallic V-block of such size that the body of the resistor does not extend beyond the extremities of the block. The resistor leads shall be so positioned that the distance between the resistor lead and any point of the V-block is not less than the radius of the resistor minus the radius of the lead wire.

b. Initial measurement: DC resistance shall be measured as specified in 4.7.4.

c. Magnitude of test voltage: Sine wave test potential of 900 volts rms, except that for styles RSC50, RSC55, and RSC60 the potential shall be 450 volts.

d. Nature of potential: An ac supply at commercial-line frequency (not more than 100 hertz) and waveform.

e. Rate of application of test voltage: Greater than 100 volts per second.

f. Duration of application of test voltage: One minute.

g. Points of application of test voltage: Between the resistor terminals connected together and the V-block.

h. Measurement after test: DC resistance shall be measured as specified in 4.7.4.

i. Examination after test: Resistors shall be examined for evidence of flashover, mechanical damage, arcing, and insulation breakdown.

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4.7.10.2 Barometric pressure (reduced). Resistors shall be tested in accordance with method 105 of MIL-STD-202. The following details and exceptions shall apply.

a. Method of mounting: As specified in 4.7.10.1a.

b. Initial measurement: DC resistance shall be measured as specified in 4.7.4.

c. Test condition: D (100,000 feet).

d. Test voltage during subjection to reduced pressure: 300 volts rms, except that for styles RSC50, RSC55, and RSC60 the potential shall be 200 volts.

e. Nature of potential: As specified in 4.7.10.1d.

f. Duration of test: One minute.

g. Points of application of test voltage: As specified in 4.7.10.1g.

h. Final measurement: DC resistance shall be measured as specified 4.7.4.

i. Examinations after test: As specified in 4.7.10.1i.

4.7.11 Insulation resistance (see 3.17). Resistors shall be tested in accordance with method 302 of MIL-STD-202. The following details and exceptions shall apply:

a. Test condition: 100 volts.

b. Special preparations: As specified in 4.7.10.1a.

c. Points of measurement: Between the resistor terminals connected together and the V-block.

4.7.12 Resistance to soldering heat (see 3.18). Resistors shall be tested in accordance with method 210 of MIL-STD-202. The following details and exceptions shall apply:

a. Measurement before test: DC resistance shall be measured as specified in 4.7.4.

b. Special preparation of specimen: Both leads shall be dipped in RMA flux in accordance with MIL-F-14256, and then dipped into solder, both for 5 ±1/2 seconds. The bath shall be maintained at 260°C ±5°C. The part shall be immersed to within .075 ±.025 inch of the body.

c. Test condition: C (260°C ±5°C, 10 +2 seconds). A board with a maximum area of nine square inches shall be used, and the leads shall not be cut.

d. Post test conditioning: The resistors shall be inserted in a vapor degreasing apparatus containing boiling 1-1-1 trichlorethane for 1 1/2 +1/2, -0 minutes. The parts shall then be cleaned with isopropyl alcohol.

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e. Measurement after test: After completion of the cleaning process and following a minimum 3-hour cooling period, the dc resistance shall be measured as specified in 4.7.4.

f. Examination after test: Resistors shall be examined for evidence of mechanical damage.

4.7.13 Moisture resistance (see 3.19). Resistors shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: Soldered by their leads to rigid mounts or terminal lugs. The spacing of the mounts or terminal lugs shall be such that the length of each resistor lead is approximately 3/8-inch when measured from the edge of the supporting terminal to the resistor body. One half of the sample units shall be covered with a V-shaped metal strap whose width is equal to that of the resistor body as indicated on figure 2. The strap shall be made of corrosion-resistant metal and shall be kept in contact with the resistor body by supporting the body as indicated on figure 2, with a nonconducting, noncorrosive support whose width is less than that of the body and which will not act as a moisture trap. The mounting straps may be individual for each resistor or continuous for all resistors.

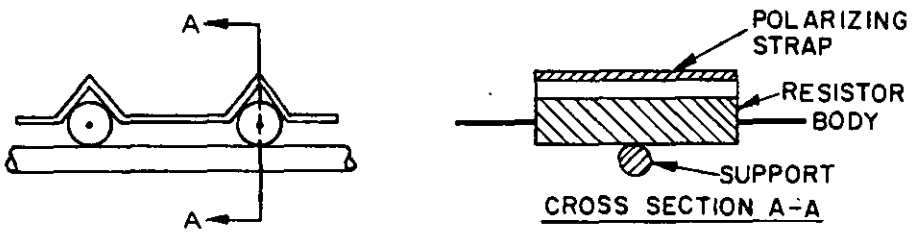


FIGURE 2. Mounting straps for moisture-resistance test.

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b. Initial measurement: Immediately following the initial drying period, dc resistance shall be measured as specified in 4.7.4.

c. Polarization and loading voltage: The resistance value selected shall be divided equally as possible for polarization and load.

(1) Polarization voltage: During steps 1 to 6 inclusive, a 100-volt dc potential shall be applied only to those resistors which have polarizing strap. This potential shall be applied with the positive lead connected to the resistor terminals tied together, and the negative lead connected to the polarizing straps.

(2) Loading voltage: During the first 2 hours of steps 1 and 4, a dc test potential equivalent to 100 percent rated wattage, but not exceeding the maximum rated voltage shall be applied to those resistors which do not have the polarizing strap specified in 4.7.14a.

d. Subcycle: Steps 7b shall not be applicable. Step 7a shall be performed during any five of the first nine cycles only.

e. Final measurements: Upon completion of step 6 of the final cycle, the resistors shall be held at the high-humidity conditions and a temperature of 25°C ±2°C for a period of 1-1/2 to 3-1/2 hours. The same straps used for polarizing the resistors may also be used for the dielectric withstanding voltage and insulation resistance tests. Resistors shall be removed from the chamber and within 1/2 hour, without any additional handling, the dc resistance, dielectric withstanding voltage, and insulation resistance shall be measured in that order as specified in 4.7.4, 4.7.10.1, and 4.7.11, respectively. The sample units shall not be subjected to forced circulation air during the tests.

f. Examination after test: Resistors shall be examined for evidence of mechanical damage.

4.7.14 Resistance noise (see 3.20). The resistance noise of resistors shall be performed using a radiometer component linearity tester type CLT1A or equivalent. A generalized schematic of the test equipment is shown on figure 3. The amount of power dissipated by the resistor shall be one half of its rated power rating or 0.25 watt, whichever is less.

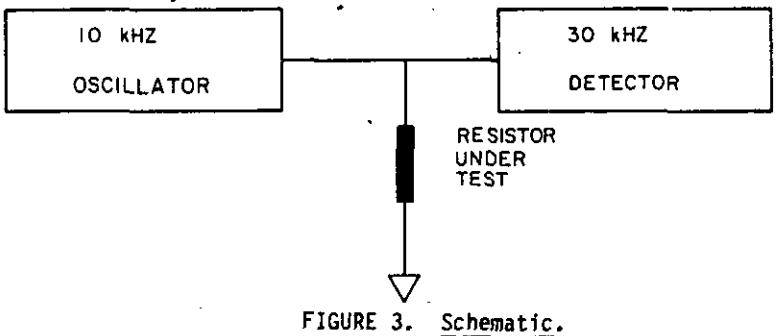


FIGURE 3. Schematic.

The third harmonic index (THI) in dB shall be calculated as follows:

$$THI\ (dB) = 20 \log V_3/V_1$$

where: V_1 = rms fundamental voltage amplitude in volts
V_3 = rms third harmonic voltage amplitude in microvolts

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4.7.15 Destructive physical analysis (see 3.21). Resistors shall be examined as specified in appendix B of this document.

4.7.16 Outgassing (see 3.22). Resistors shall be tested in accordance with ASTM E-595. The following details and exceptions shall apply. Samples of any materials used shall be cut into pieces having 0.38 inch (246 mm) maximum dimension. A sample of 100 to 300 milligrams of material shall be weighed and then placed in a vacuum chamber under the following conditions:

Pressure - - - - - - - - - -	10^{-6} torr or less.
Temperature of specimen- -	Maximum operating temperature of device.
Exposure time - - - - - -	24 hours.

During this exposure, a collector plate 9.7 inches (246.4 mm) in diameter shall be held at 25°C ±1°C near the sample. Immediately upon removal of the specimen and collector plate from the chamber, they shall be weighed. The Total Mass Loss (TML) is sample mass before exposure minus the sample mass after exposure divided by the initial sample mass (expressed in percent change). The Volatile Condensable Material (VCM) is the increase in the weight of the collector plate divided by the mass of sample (expressed in percent).

5. PACKAGING

5.1 Packaging requirements. Resistors shall be prepared for delivery in accordance with MIL-R-39032.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Title, number, and date of the applicable detail specification, and the complete PIN designation (see 1.2.1 and 3.1).

d. Levels of preservation and packaging, and packing, and applicable marking (see section 5).

e. Requirement for notification of exceeding change of resistance in excess of that specified in 6.5.

f. Marking requirement for failure rate level, temperature characteristic and tolerance.

g. Lead length: Specify 1.00 +.625, -0.000 inch for tape and reel packaging. If not specified, the lead length will be in accordance with the detail specification (see 3.1).

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6.2 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time of award of contract, qualified for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Governmnt tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Electronics Command; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-E), 1057 Wilmington Pike, Dayton, Ohio 45444.

6.3 Part or Identifying Number (PIN). The PIN shall be structured in accordance with 1.2.1.

6.4 Subject term (key word) listing.

Resistor
Resistor film
Resistor fixed
Space level quality

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 19
NASA - NA

Review activities:
Army - AR
Navy - AS, OS
Air Force - 11, 17, 80, 85
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC

Preparing activity:
Air Force - 19

Agent:
DLA - ES

(Project 5905-1106)

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APPENDIX A

PROCEDURES FOR THE PRODUCT AND QUALITY AUDIT

10. SCOPE

10.1 Scope. The appendix provides procedures to the Product and Quality Audit team for the auditing of manufacturers supplying to this specification for certification of manufacturing facilities. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. INSTRUCTIONS

30.1 Baseline document. The manufacturer shall complete the baseline document and submit the required information to the qualifying activity prior to scheduling the Product and Quality Audit. It is not intended that the manufacturer submit proprietary information. The baseline document shall contain the following information, as a minimum:

a. Detailed flow charts indicating the following:

   (1) All significant process steps and measurement and inspection points by descriptive name and document number.

   (2) All raw material and interim and finished part quality control points by descriptive name, document number, and location in the processing sequence. Documents shall include forms and travelers.

   (3) All raw material incoming inspection, storage, and processing steps by descriptive name, document number, and location in the processing sequence. Documents shall include forms and travelers.

   (4) The identification of all critical (significant to end product properties) manufacturing and inspection steps as determined by the manufacturer in conjunction with the audit team at the time of baseline document approval.

b. A listing of the documents identified in the flow charts indentifying current revision letter, revision date and classification (i.e., critical, proprietary).

c. Blank copies of all current forms and travelers cited in the flow charts.

d. Copies of quality control documents detailing sampling plans used in inspections cited on flow charts and the quality control inspector training.

e. Procedures to control documentation, documentation changes, and distribution of the documents or document changes to all required personnel.

f. Procedures for raw material control for acquisition, incoming inspection, storage, withdrawal from inventory, and shelf life control.

g. Details of part design delineating dimensions, tolerance, materials, and internal part numbers.

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h. Procedures for controlling computer software including basic flow chart for program logic, current revision numbers, program access controls, and storage of part revisions of programs.

i. A complete list of process and test equipment utilized.

j. Organizations chart of key personnel and their relations.

k. List of raw materials and approved/qualified supplies.

l. List of approved outside test and service laboratories used.

m. Procedures for statistical process control and failure analysis.

30.1.1 Changes to baseline. The audit team shall compare the latest baseline document to the previous baseline document to assure all changes have been submitted to the qualifying activity and have been approved.

a. Minor changes. Any subsequent departure from the approved flow chart (i.e., additions or deletions) or changes to the revision status of those processes or inspections are allowed, provided they do not violate the changes prohibited by section B (major changes). These minor changes shall be described in general terms to the audit team during subsequent audits. This action shall result in an update of the baseline document.

b. Major changes. No changes (i.e. substitutions, additions, deletions, or modifications) shall be allowed in the following areas without prior review and approval of the qualifying activity. In these, and other areas deemed major by the audit team at the time of baseline approval, partial or total product requalification may be required as determined by the qualifying activity.

(1) Significant raw materials.

(2) Qualified sources for significant raw materials.

(3) Substrate preparation.

(4) Procedure for metallization or deposition.

(5) Element stabilization.

(6) Trimming or resistance adjustment.

(7) Capping and lead attachment.

(8) End cap plating and cleaning.

(9) Coating process.

(10) Encapsulation or sealing process.

(11) Group A, B, or C inspections.

30.2 Audits. An initial audit and subsequent periodic audits (every year or as deemed necessary) shall be performed by the qualifying activity. The audit shall include, but not be limited to, all of the items on the checklist (plus MIL-STD-790) and any GIDEP ALERTS issued against the manufacturer at the facility of interest. The results of any additional items that are assessed shall be added to the checklist for record purposes.

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30.2.1 Duties of the audit team.

a. The audit team shall review the manufacturer's facility and the proposed baseline documentation for adequacy, accuracy, and completeness during the audit of the manufacturer. The proposed baseline documentation should be used as a baseline/reference during the course of the audit.

b. The audit team shall verify that the test measuring procedures and equipment are sufficently accurate for the required test.

c. The audit team shall typically sample check twelve months of records for the items being certified. In addition, the location of the equipment and the thoroughness and adequacy of the operating procedures and instructions shall be checked. Whenever practical, operations utilizing critical equipment and procedures shall be witnessed.

Check - item	Inspection	Satisfactory
a. Verification of MIL-STD-790 requirements.		
b. Flow chart (1) Key processes identified with revision. (2) Key inspection points identified with revision. (3) Quality control points identified with revision. (4) Key raw material control documents identified with revision. (5) Critical steps identified. (6) Nonproprietary documents reviewed. (7) Test and measurements points identified with revision.		
c. Baseline: Shall describe, as a minimum, the following: (1) Resistor design by part number indicating the following with the necessary tolerance defined: End cap alignment Body length Body diameter Shell end metallization (2) Documents listing suppliers qualified for all raw materials.		
d. Lot control plan (see 4.6.1.1).		
e. Traceability plan.		
f. Training plan for all personnel.		

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APPENDIX B

PROCEDURES FOR DESTRUCTIVE PHYSICAL ANALYSIS (DPA)

10. SCOPE

10.1 Scope. This appendix provides the methods and detailed requirements for DPA of metal film resistors. This appendix is a mandatory part of the specification contained herein is and intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. EQUIPMENT AND PROCEDURES.

30.1 EQUIPMENT.

30.1.1 Sample preparation. A mechanical or chemical process shall be used to remove the outer insulating materials. This process shall cause minimal damage to the internal resistor structure.

30.1.2 Optical equipment. A binocular microscope with a minimum capability of 100x magnification and on integral light source or fiber optic light ring shall be used.

30.2 Resistor, fixed, film, styles RSC50, 55, 60, and 65.

30.2.1 Method

30.2.2 External visual. Examine resistors at 30x minimum magnification for cracks on epoxy coating or glass seal, flaws on the leads (nicks, cuts, crushing, or exposure of the base metal) and marking, and nonconformance with configuration requirements.

30.2.3 Hermeticity. Conduct seal test on each sample, when applicable, after all other tests have been performed and just prior to sample preparation for internal inspection. Seal test shall be performed in accordance with the requirements of the procurement specification.

40. SAMPLE PREPARATION

40.1 Conformally coated resistors. Conformal coating can be mechanically or chemically removed. When mechanical means are employed, care must be taken not to induce any rejectable defect on the resistor itself. When chemically stripping the coating, use a process and material which dissolves the coating and exposes, but not attacks, the metal film, core, end terminations, leads, and welded connections, and which does not discolor or stain any surface. As required, perform SEM inspection and check for corrosion.

40.2 Hermetically sealed resistors.

40.2.1 Conformally coated, ceramic body. Strip resistor's conformal coating in accordance with 40.1, then examine the hermetic packaging for any signs of contamination, cracks on the ceramic sleeve, and pits or voids on the solder seal. Open the ceramic outer packaging by using a small lathe machine to expose the metallized resistor, being careful not to introduce any contamination. Prepare the lathe by cleaning both jaws and tool bit with alcohol to remove the oils. Carefully place the resistor between the jaws so that one end protrudes. (Note: Do not close the jaws too tightly.) Cut the metal end cap, keeping the lead attached, just beyond the corner radius. Repeat this procedure for the other end, then slide the ceramic sleeve out to expose the internal resistor.

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40.2.2 Glass body. Scribe the glass sleeve with a diamond wire saw, then place the resistor on a lint-free tissue, and fold the tissue over. Lightly tap the device with a very small hammer, or similar device, until the glass breaks and the glass chips can be removed. Care must be exercised in performing this procedure in order not to induce damage to the internal resistor.

40.3 Removal of internal end caps. The end caps must be removed to inspect for any sign of corrosion, blistering, peeling away of the plating material, or metal film under the caps. This procedure shall be performed after the internal visual inspection has been completed. Roll the internal resistor between two clean flat hard surfaces. An aluminum pad and a sheet of glass has been used with success. After rolling several times, the end caps should fall off from the resistor core. Typically, the caps come off one at a time. When rolling the remaining cap, care must be taken to ensure that damage is not induced to the exposed end. Inspect caps and ends of resistor element at 30x minimum magnification. Any corrosion, peeling or blistering of the plating or metal film under the caps shall be cause for rejection.

50. DESTRUCTIVE PHYSICAL ANALYSIS

50.1 Data records. DPA findings that deviate from configuration and other requirements shall be documented as defects.

50.2 Evaluation criteria. When the DPA is conducted as a lot conformance test, the production lot shall be rejected if one or more of the DPA samples exhibit any of the defects listed below. All exposed inner surfaces of each resistor shall be examined at 30x minimum magnification for the following characteristics:

a. Cracked or chipped core.

b. End cap misalignment greater than 3 degrees.

c. Weld splatter at lead-to-end cap termination, or cracks in weld joint.

d. Cracked, split, or holes (from welding operation) on end caps.

e. Corrosion, lifted, blistered, or missing plating material on the end caps; discoloration due to welding is not cause for rejection.

f. Foreign material or contamination on the metal film.

g. Any particle other than encapsulation material seated on the spiral cut.

h. Feathering of the metal film due to laser trimming that extends to more than 50 percent of the width of the trim.

i. Damage to resistnace element, such as overtrimming, scratches, or voids, which reduces the width of any of the metallized turns to less than 50 percent of the original design value.

j. Number of spiral turns different between samples having similar resistance values, up to 5 percent or 1 turn, whichever is greater.

k. Evidence of heavy stain or corrosion on resistance element. Discoloration of tantalum-based resistors due to thermal stabilization or laser trimming shall not be cause for rejection.

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l. Metal film lifting, peeling, or blistering as observed visually or determined by a cellophane tape test.

m. Intermediate coating, when used, that does not extend to 1/2 the length of the skirt of the end caps.

n. Discoloration, foreign material, bubbles or pinholes on the intermediate coating, or coating that is peeling away.

o. Discolored or hollow core.

p. Excessively deep cuts due to laser trimming, i.e., laser trim cuts of more than 0.007 inch deep for thick films or 0.0005 inch deep for thin films.

q. Uncut material remaining after a laser scribe due to "skipping" of the laser beam.

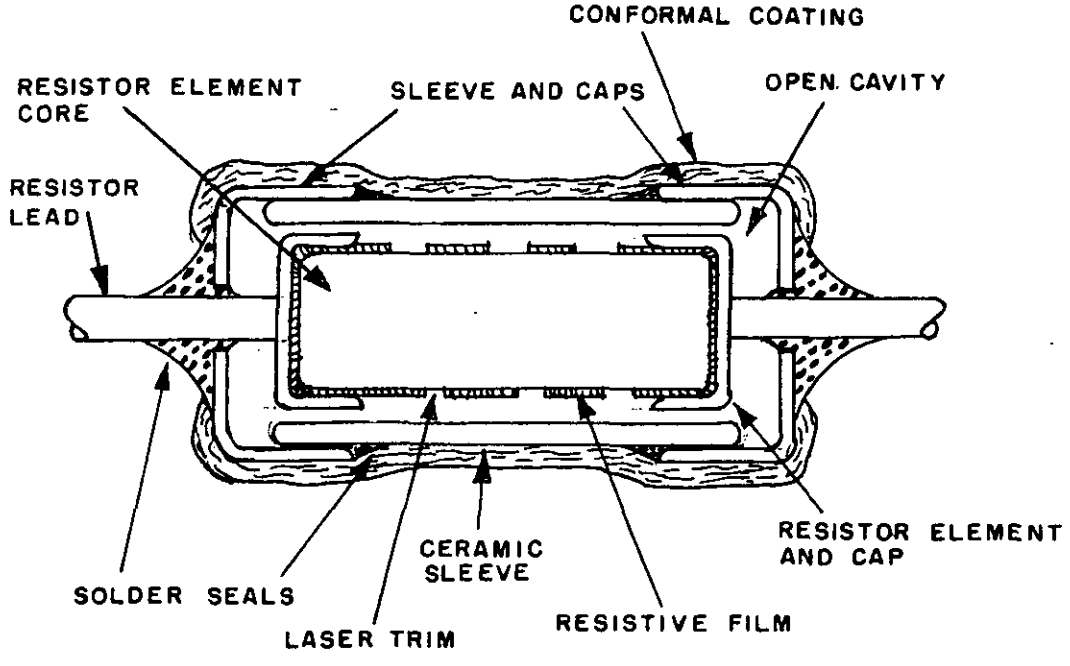


FIGURE 4. Typical hermetically sealed film resistor, RSC style.

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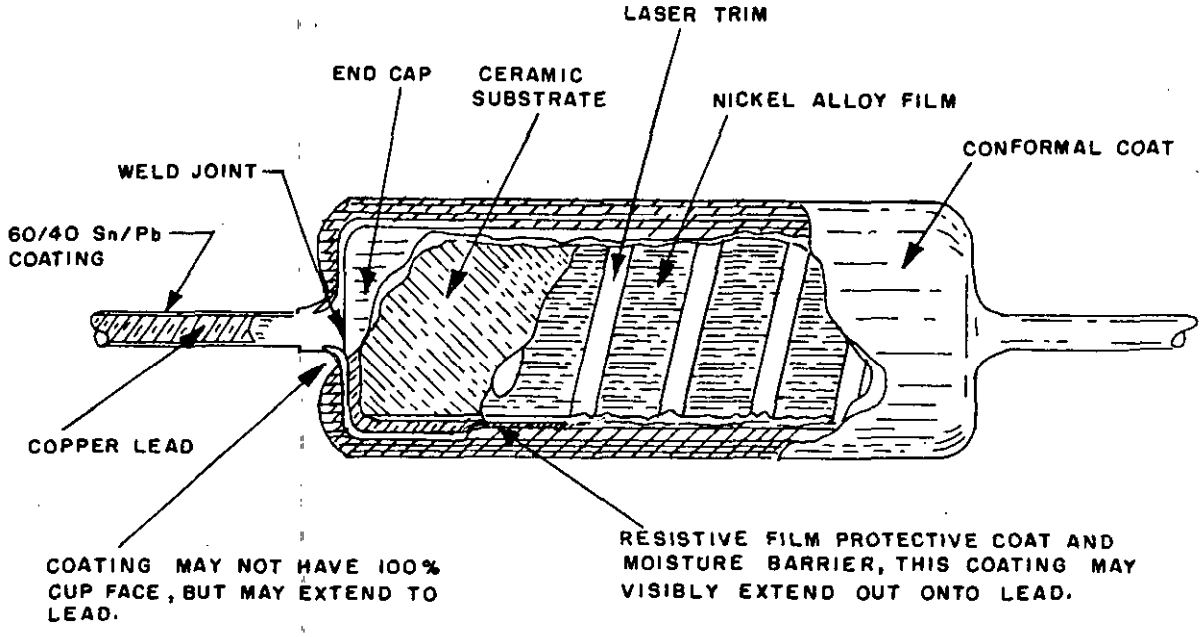


FIGURE 5. Typical nonhermetically sealed film resistor, RSC style.

60. Resistor, fixed, film, style RSC90.

60.1 Method.

60.2 External visual. Conduct external visual examination as in 30.2.2.

60.3 Terminal strength. A terminal strength test shall be performed on 1/3 of the samples in accordance with the procurement specification.

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60.4 Sample preparation. Either of the following methods may be used in opening (exposing) resistor for internal examination:

a. Chemical decapsulation. The case may be chemically removed using a process and material which dissolves the case and the underlying coatings and exposes, but not attacks, the resistor element, substrate, terminations, leads, and welded connections, and which does not discolor or stain any surface. The solvent used shall be identified and recorded in the report.

b. Mechanical decapsulation.

(1) Gently hold the sample at the two narrow sides parallel to the leads. Slowly grind off the flat (wide) side of the case with a 180, or finer, grit paper until the plastic is thin enough to break off with pointed tweezers or when the rubber coating starts to show through.

(2) Repeat step 1 for the other flat side. When only the four narrow sides have the plastic case material still attached to them, gently break these off with a pair of pointed tweezers.

(3) The rubber film surrounding the substrate package can be easily removed with the tweezers. However, the moisture barrier coating (varnish) under the rubber may not be as easy to take out. Carefully and slowly scrape the varnish with the toothpick and tweezers making sure that no rejectable defects are introduced to the resistor element.

70. DESTRUCTIVE PHYSICAL ANALYSIS

70.1 Data records. DPA findings that deviate from configuration and other requirements shall be documented as defects.

70.2 Evaluation criteria. When the DPA is conducted as a lot conformance test, the production lot shall be rejected if one or more of the DPA samples exhibit any of the defects listed below. All exposed inner surfaces of each resistor shall be examined at 30x minimum magnification for the following characteristics:

a. Cracked or chipped substrate or film. However, small chips on the edges of the substrate are permissible as long as they do not extend under the film.

b. Substandard welding, such as:

(1) Weld print missing the film pads by more than 20 percent.

(2) Half of each weld print is outside the lead outline.

(3) Lead is over active grid lines.

(4) Weld splatter.

(5) Cracked paddle or the flat welded portion of lead terminal.

(6) Torn ribbon connection.

(7) Epoxy dot material running under or into weld connections.

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c. Cracked, nicked, or dented lead terminal.

d. Resistor images which indicate hot spot possibility by a 20 percent reduction of path-metal.

e. Lifting or blistering of resistor metal.

f. Heavy scratches or gouges in active area of resistor element.

g. Any particle large enough to bridge active grid lines.

h. Three or more particles of any size that are detectable under 30x magnification.

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